Correcting for AI-Screened Poverty Misclassification in Child Health Regressions: A Method-of-Moments Approach and Its Small-Sample Instability, Applied to Ethiopian Household Panel Data

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Abstract

A standard fix exists for the bias introduced when a noisy screen (an LLM classification, a proxy means test) stands in for a true poverty indicator in a health regression: the Rogan-Gladen method-of-moments correction, which rescales the naive coefficient using a validation sample's estimated sensitivity and specificity. We show this fix has a failure mode that does not show up in typical validation-sample sizes and is easy to miss if you only check that the correction "works" on average. Using real household consumption and child height-for-age (HAZ) data from the World Bank Ethiopia Socioeconomic Survey (2011–2015, n = 4,853 household-waves), we simulate a poverty screen of fixed, known accuracy (80%) and draw 20 independent validation samples of a realistic small-survey size (n = 38). One in twenty draws happens to estimate classifier skill (TPR+TNR−1) at just 0.114 – a plausible sampling outcome, not a broken classifier – and the correction, applied mechanically, returns a coefficient of −34,545: five orders of magnitude from any sensible value, despite every draw sharing the same true 80%-accurate screen. We trace the mechanism term by term, show it compounds a small denominator in the correction with a boundary-clipped prevalence estimate, and connect it formally to weak-instrument asymptotics, where a structurally identical ratio estimator is known to misbehave when its denominator is small relative to its own sampling variance. On the headline, better-behaved specification, the same correction performs as expected: the naive estimator attenuates the poverty-HAZ association – estimated, here and throughout, using correctly measured rather than screened poverty status – by 43% (−0.204 vs. a benchmark of −0.357), and the correction recovers 94% of that gap (−0.366). We propose a practical minimum-skill threshold below which a corrected estimate should be flagged rather than trusted, and, to our knowledge, provide the first application of this correction to a real, independently-measured poverty indicator and real health outcome, rather than a simulated ground truth.

JEL Classification: C81, C83, I32, O12

Keywords: measurement error; misclassification bias; large language models; proxy means testing; child health; Ethiopia; Rogan-Gladen correction

1 Introduction

Administering a full household consumption module – the standard basis for a poverty line in Living Standards Measurement Study (LSMS)-style surveys – is the single most expensive and time-consuming part of collecting welfare data. This has motivated a decades-long literature on proxy means testing: predicting welfare from a short list of easily observed household characteristics rather than measuring it directly (Grosh and Baker, 1995; Brown et al., 2018). The rise of large language models adds a new, faster variant of the same idea: an LLM can classify a household as poor or non-poor from a brief description – a field note, a chatbot-mediated intake conversation, an enumerator's shorthand – without a formal proxy-means-test regression at all (Gilardi et al., 2023). Humanitarian and development organizations are already exploring LLMs for exactly this kind of rapid classification of qualitative field data (Marston et al., 2026). Work in this vein, Gilardi et al. (2023) included, evaluates classification quality on its own terms – accuracy, agreement with human coders, cost. This paper is not that kind of study; it asks what happens one step further downstream, once a classifier of some given, known accuracy has already been used to generate a regressor.

This paper asks a narrower, more mechanical question than whether such a screen is a good idea in general: *if* you use one, and its output enters a regression as an independent variable instead of the true poverty status, how much bias does that introduce, and can you correct for it economically – with a validation sample, not a second full survey? This is a classical measurement-error question with a known answer in principle (Aigner, 1973; Bollinger, 1996): a binary regressor observed with non-differential classification error attenuates the estimated coefficient toward zero, and the attenuation is exactly correctable if the classifier's sensitivity (true positive rate, TPR) and specificity (true negative rate, TNR) are known, via the method-of-moments correction of Rogan and Gladen (1978).

What is missing from the existing demonstrations of this correction – including two companion pipelines we built alongside this one, applied to a simulated financial-sentiment outcome and a fully synthetic outcome with a planted effect size – is an application where *both* the mismeasured regressor's true value and the downstream outcome are independently real, measured quantities, rather than one or both being generated by the analyst to have a known answer. Here we use the World Bank's Ethiopia Socioeconomic Survey (ESS), a genuine household panel with real per-adult-equivalent consumption (the basis for our true poverty indicator) and real child height-for-age z-scores (HAZ, the basis for our outcome), so that the "benchmark" regression we correct toward is itself an empirical estimate with its own sampling uncertainty, not an oracle.

We make two contributions, and lead with the one we think is more useful for a practitioner rather than the one we set out to demonstrate. First, we show that this correction's reliability depends sharply on the size of the validation sample used to estimate TPR and TNR – not just asymptotically, but sharply enough that at realistic small-survey validation sizes ($n = 38$), the correction diverges by five orders of magnitude in roughly one out of twenty random splits, purely from sampling noise in the skill estimate, with the true screen accuracy held fixed throughout. We characterize this instability as a function of the estimated classifier skill (TPR+TNR$-$1), trace its mechanism term by term rather than only demonstrating it empirically, connect it formally to weak-instrument asymptotics, and propose a practical minimum-skill screening rule. Second, we quantify the misclassification-bias-and-correction pipeline end-to-end on a real pairing of screen and outcome: a simulated screen with 80% target accuracy (TPR$= 0.76$, TNR$= 0.81$ realized on our validation split) attenuates the poverty-HAZ gradient by 43%, and the correction recovers 94% of the resulting gap.

Section 2 reviews the measurement-error mechanics and the proxy-means-testing literature this sits within. Section 3 describes the ESS data, the (simulated) screening task, and why the correction requires it to be non-circular. Section 4 states the three estimators and the identifying assumption. Section 5 reports the headline result and the two sensitivity analyses. Section 6 draws out the practical implication and situates this relative to the companion synthetic-data pipelines. Section 7 concludes.

2 Background

2.1 Proxy-based poverty screening as a live practice

Proxy means testing – predicting household welfare from easily observed correlates rather than measuring consumption directly – has been used for social-program targeting since at least Grosh and Baker (1995). Brown et al. (2018) assess proxy-means-test performance across nine African countries and find that, while standard PMTs filter out many non-poor households, they also exclude a substantial share of the genuinely poor – i.e., real, non-trivial misclassification is the norm in fielded proxy targeting systems, not an edge case. LLM-based screening from unstructured text extends the same underlying idea (predicting welfare from cheap correlates) to a different feature set – language rather than a fixed covariate list – and researchers are already applying it to coding qualitative humanitarian data more broadly (Marston et al., 2026). Whatever the feature set, the same measurement-error mechanics apply once a program uses the screen's output as if it were the true classification.

2.2 Attenuation from a mismeasured binary regressor

Let $D^* \in \{0, 1\}$ denote true poverty status, with prevalence $q \equiv \Pr(D^* = 1)$, and let $D \in \{0, 1\}$ denote its imperfect classification, with sensitivity $\Pr(D = 1 \mid D^* = 1) = \text{TPR}$ and specificity $\Pr(D = 0 \mid D^* = 0) = \text{TNR}$. A terminological note before proceeding, since the word recurs throughout this paper: "true" here, and in "true prevalence," "true screen accuracy," and similar phrases below, means *correctly measured* – D^* is poverty status free of classification error, not an assertion that poverty causally determines the outcomes we regress it on. The two questions are unrelated, and the "What the correction does not fix" paragraph later in this section returns to the causal question on its own terms. Write $\mathrm{FPR} \equiv 1 - \mathrm{TNR}$ and define

$$\kappa \equiv \mathrm{TPR} + \mathrm{TNR} - 1 = \mathrm{TPR} - \mathrm{FPR}, \tag{1}$$

the classifier's discriminating skill above chance – zero for a coin flip, one for a perfect classifier, and, not coincidentally, exactly Youden's J statistic from the diagnostic-testing literature. Suppose the structural relationship of interest is $Y = \beta_0 + \beta_1 D^* + \varepsilon$ with $E[\varepsilon \mid D^*] = 0$, so that $\mathrm{Cov}(D^*, Y) = \beta_1 \mathrm{Var}(D^*) = \beta_1 \, q(1-q)$. Misclassification is *non-differential* if $D \perp Y \mid D^*$ – the screen's error depends on the truth D^* but not, additionally, on the outcome itself.

Why the naive coefficient attenuates: a derivation, not an assertion. It is not enough to assert that regressing Y on D shrinks β_1 toward zero (Aigner, 1973); the mechanism is worth deriving explicitly, because the same derivation is what motivates the exact functional form of the correction in Equation 6 below. By the law of total covariance,

$$\mathrm{Cov}(D, Y) = \underbrace{E\big[\mathrm{Cov}(D, Y \mid D^*)\big]}_{=0 \text{ under non-differential error}} + \mathrm{Cov}\big(E[D \mid D^*],\, E[Y \mid D^*]\big).$$

The first term vanishes exactly because $D \perp Y \mid D^*$. For the second term, note $E[D \mid D^*] = \mathrm{FPR} + \kappa D^*$ (it equals TPR when $D^* = 1$ and FPR when $D^* = 0$, which this expression interpolates linearly), and $\mathrm{Cov}(D^*, E[Y \mid D^*]) = \mathrm{Cov}(D^*, Y)$ by the same total-covariance identity applied to D^* itself. Since FPR is a constant, it drops out of the covariance, leaving the central identity this paper builds on:

$$\mathrm{Cov}(D, Y) \;=\; \kappa \cdot \mathrm{Cov}(D^*, Y) \;=\; \kappa \, \beta_1 \, q(1-q). \tag{2}$$

The observed covariance is the true covariance scaled down by the classifier's skill κ – nothing more exotic than that. But the naive OLS coefficient is not this covariance alone; it is $\hat{\beta}_1^{\text{naive}} \xrightarrow{p} \text{Cov}(D, Y)/\text{Var}(D)$, and $\text{Var}(D) = q^*(1 - q^*)$ where the *observed* positive rate $q^* = \Pr(D = 1) = \text{FPR} + \kappa q$ generally differs from the true prevalence q. Combining these,

$$\hat{\beta}_1^{\text{naive}} \xrightarrow{p} \beta_1 \cdot \kappa \cdot \frac{q(1-q)}{q^*(1-q^*)}. \tag{3}$$

Two distinct sources of attenuation are visible here, not one: the skill discount $\kappa < 1$ acting directly on the covariance, and a second scaling by the ratio of true-to-observed prevalence variance, which is generally different from κ itself and need not even move the estimate in the same direction on its own (though for the empirically relevant case of $\text{TPR}, \text{TNR} > 0.5$ and κ not too close to one, it typically compounds the shrinkage). Equation 3 is the binary-regressor special case of the classical measurement-error attenuation result (Aigner, 1973); Bollinger (1996) derives bounds on β_1 for the case where TPR and TNR are unknown rather than estimated from a validation sample, as they are here.

From attenuation to correction: Equation 6 is not a separate formula. Rearranging Equation 2 for β_1 gives

$$\beta_1 = \frac{\text{Cov}(D, Y)/\kappa}{q(1-q)}. \tag{4}$$

The numerator is directly computable from the observed data once κ is known (from a validation sample): dividing the observed covariance by the skill discount exactly undoes the shrinkage in Equation 2. The obstacle is the denominator, $q(1 - q)$, which requires the *true* prevalence q – and q is precisely what is unobserved when only the noisy classification D is available. This is exactly the gap the Rogan-Gladen prevalence correction (Rogan and Gladen, 1978) fills, by inverting $q^* = \text{FPR} + \kappa q$ for q:

$$\hat{q} = \frac{q^* - \text{FPR}}{\kappa}, \tag{5}$$

a method-of-moments estimator that is consistent for q as the validation-sample estimates of TPR and TNR converge to their population values. Substituting $\hat{q}$ for the unobserved q in Equation 4 yields the coefficient correction used throughout this paper:

$$\hat{\beta}_1 = \frac{\text{Cov}(D, Y)}{\kappa \cdot \hat{q}(1 - \hat{q})}. \tag{6}$$

Equation 6 is therefore not an independent rescaling rule alongside Equation 5 – it is the unique estimator obtained by plugging the Rogan-Gladen prevalence estimate into the inverted attenuation identity, Equation 4. Consistency follows immediately: since $\hat{q} \xrightarrow{p} q$, Equation 6 converges to $\text{Cov}(D, Y)/[\kappa\, q(1 - q)] = \beta_1$ by Equation 2.

A numerical check. At the realized validation-sample rates in our main specification (TPR$=0.762$, TNR$=0.813$, so $\kappa = 0.575$) and a true within-wave poverty prevalence of $q \approx 0.333$ by construction, Equation 3 predicts the naive estimator should retain $\kappa \cdot q(1 - q)/[q^*(1 - q^*)] \approx 0.575 \times 0.222/0.235 \approx 54\%$ of the true coefficient – a 46% predicted attenuation. Section 5 reports an empirically *observed* attenuation of 43% on the study sample. The two numbers are close but not identical, as expected: Equation 3 is a population-level probability limit, while the study sample is finite ($n = 3{,}883$) and its realized prevalence is not exactly 1/3. We view this agreement as an informal specification check – the closed-form attenuation formula derived above predicts, to a reasonable approximation, what the empirical regressions in Section 5 independently produce.

What the correction does not fix. Equations 5–6 correct one specific problem: bias introduced by using a noisy D in place of D^*. They say nothing about whether β_1 itself – the coefficient on the *true* poverty indicator D^* – has a causal interpretation. This is a property of the benchmark regression itself, independent of any screen: even the benchmark in Section 4, which uses real, unscreened poverty status, is a bivariate association with no controls, and household poverty and child HAZ plausibly share common causes that this design does not adjust for, whether or not an AI screen, or any screen, is anywhere in the picture. Geographic remoteness is a concrete, real example: a household further from markets and health facilities, or at an elevation and rainfall regime associated with weaker agricultural yields, is plausibly both poorer and has a child with lower HAZ through pathways that do not run through household poverty at all – worse water and sanitation infrastructure, or reduced access to diverse foods and health services, rather than income as such. (These same geographic variables happen also to feed the field note in Section 3.4, since they are covariates already present in the panel – a coincidence of data availability, not the reason they confound.) If geography confounds in this way, β_1 in every one of our three estimators is confounded by it in the same direction and by a similar magnitude, and the correction we study faithfully recovers this confounded association, not a causal effect of poverty on child health. Reverse causality is a further, distinct concern the correction is equally silent on: a chronically sick child raises a household's medical spending and can depress its measured consumption within the same wave, which would bias β_1 even under perfect measurement of D^*. We do not attempt to resolve either concern here – doing so would require an identification strategy (e.g., household or village fixed effects, or an instrument for poverty) orthogonal to the measurement-error problem this paper studies – and we return to this scope limitation in Section 6.

Both corrections above share the denominator κ. As Equations 5–6 make algebraically plain, and as Section 5.2 demonstrates is not merely a theoretical curiosity, both are numerically unstable as $\kappa \to 0$ – and κ is itself *estimated* from a finite validation sample, not a known constant. Section 5.2 characterizes exactly how this instability propagates through Equation 5 into Equation 6, and draws out its structural similarity to a well-studied problem in a different corner of econometrics: weak-instrument asymptotics (Staiger and Stock, 1997).

3 Data

3.1 The Ethiopia Socioeconomic Survey

We use the same World Bank Ethiopia Socioeconomic Survey (ESS) household panel described in the companion rainfall-shocks study in this portfolio: Panel I, waves 2011–12, 2013–14, and 2015–16, approximately 3,969 households re-interviewed across waves. From the processed household-wave panel we use two real, independently meaningful variables: `nom_totcons_aeq` (nominal total consumption per adult equivalent, the standard LSMS-ISA welfare aggregate) and `mean_haz` (household-mean child height-for-age z-score). HAZ for each child is computed as the child's observed height in centimeters minus the age- and sex-specific median height from the WHO 2006 Child Growth Standards (WHO Multicentre Growth Reference Study Group, 2006), divided by that reference group's standard deviation; `mean_haz` then averages this z-score across all children under five in the household-wave. Age in months, the input the WHO reference table is indexed by, requires converting each child's recorded birthdate from the Ethiopian calendar (13 months per year, offset roughly 7–8 Gregorian years behind) to the Gregorian calendar first – an easy step to get wrong silently, since a naive same-format subtraction would misstate every child's age and therefore every HAZ value; the companion paper documents this correction in full and we reuse its output unmodified here. After dropping household-waves missing either variable, our analysis sample has $n = 4{,}853$ household-wave observations.

3.2 Data availability

The ESS/ERSS panel is public-use World Bank LSMS-ISA microdata, available at no cost from the World Bank Microdata Catalog after free registration and acceptance of the catalog's terms of use, which (standard for LSMS-ISA data) prohibit redistribution of the raw microdata itself; we therefore cannot bundle the underlying survey files with the accompanying repository, and the pipeline reads them from a local path the user supplies after downloading and registering. All code that builds the processed panel, constructs the poverty indicator and field notes, runs the simulated screen, and produces every table and figure in this paper is available in the accompanying public repository, along with a synthetic-data fallback (schema-matched, not distributionally identical) that lets the full pipeline and test suite run without access to the restricted-use files.

3.3 Constructing the poverty indicator

We define a household-wave as poor if its consumption falls in the bottom third of its *own wave's* distribution. This is a relative, within-wave definition, not an absolute poverty line: `nom_totcons_aeq` is nominal Ethiopian birr, not deflated across the 2011–2015 study period, so a fixed cross-wave birr threshold would conflate "poor by 2011 prices" with "poor by 2015 prices." Section 5 reports robustness to this threshold choice (quartile through median).

The gold standard is not noise-free either. We label D^* "true" poverty status throughout, and it is the best available measure in this dataset, but consumption aggregates constructed from household surveys carry their own well-documented measurement error – recall-period bias, seasonal consumption smoothing, and aggregation-rule sensitivity are exactly what the LSMS methodology literature on constructing consumption aggregates exists to manage (Deaton and Zaidi, 2002). This matters specifically for the validation exercise in Section 3.4: the screen's TPR and TNR are computed *against* D^*, so any noise in D^* is folded into what we report as classifier error rather than quantified separately. This is the same underlying concern documented for a related but distinct application in a companion project in this portfolio: Burke et al. (2021)'s "noisy test data" problem for satellite-based wealth prediction, where evaluating a model against a noisy observed label understates its true skill. We do not attempt to disentangle screen error from gold-standard error here, and flag this as an assumption rather than a demonstrated non-issue.

3.4 The simulated AI screen

The real ESS instrument has no free-text field, so there is no real record of an LLM (or any other) classifier's output to use directly. We instead construct, for each household-wave, a short field note from covariates already present in the panel – household size, distance to the nearest market, elevation, this year's rainfall anomaly, and reported weather shocks – framed as a community health worker's brief home-visit note, and deliberately *excluding* consumption itself, since including it would make the classification task circular. Every note follows a fixed four-sentence template (household composition, distance and elevation, rainfall anomaly, weather shocks), with each household-wave's own covariate values substituted into the same wording; nothing beyond the substituted numbers varies across notes.

We state upfront, before showing an example, what the note is and is not used for in this paper: a mock classifier is handed the note and the true label D^*, and flips the label with a probability calibrated to hit a target accuracy (80% in our main specification), *without reading the note's content* – the note is generated for realism and for the real-classifier client described below, but no result in this paper is a function of its text. The point of this design is to obtain a classifier with a known, controlled, reproducible error rate against which to evaluate the correction, not to evaluate any specific real classifier's real accuracy on this task. With that established, a household-wave with true poverty status $D^* = 1$ in our study sample generates the note:

> *"Home visit note: household has 1 child(ren) under five. It is located roughly 100 km from the nearest market, at an elevation of about 1155m. Annual rainfall this year was about 140mm above the long-run average. No weather shocks were reported this year."*

We are explicit that this whole apparatus is a controlled instrument for isolating the measurement-error mechanism, not a claim about what accuracy a real LLM would achieve reading real household descriptions; Section 6 returns to this distinction. In the notation of

Section 2.2, the consumption-based indicator from Section 3 is D^* and this classifier's output is D; Section 4 carries both forward directly.

The flip mechanism, formally. For a target accuracy $a \in (0,1)$ and a Bernoulli draw $u \sim \text{Unif}(0,1)$ independent of D^* and of the note's content,

$$D = \begin{cases} D^* & \text{if } u \le a \\ 1 - D^* & \text{if } u > a \end{cases}, \qquad \Pr(u \le a) = a \text{ regardless of the value of } D^*. \tag{7}$$

Because the flip probability $1 - a$ does not depend on D^*, this mechanism is symmetric by design: the *population* true positive and true negative rates are both exactly a ($\text{TPR} = \text{TNR} = a$), so $\kappa = 2a - 1$ exactly at the population level – a is a fixed design parameter, not itself a random variable, so no expectation operator is needed here. Any *estimate* $\widehat{\text{TPR}}, \widehat{\text{TNR}}$ computed from a finite sample of flips is, of course, a random variable and need not come out equal. The realized validation-sample asymmetry we report in Table 1 ($\widehat{\text{TPR}} = 0.762 \neq \widehat{\text{TNR}} = 0.813$, against a common target $a = 0.80$) is exactly this: finite-sample sampling noise from $n = 970$ Bernoulli draws, not an asymmetry built into the classifier. The same is true of every row of Table 3 below, each of which reports one realized $(\widehat{\text{TPR}}, \widehat{\text{TNR}})$ pair at a different target a, not the (exactly equal) population values – worth stating explicitly in both places, since Equations 5–6 use $\widehat{\text{TPR}}$ and $\widehat{\text{TNR}}$ separately and a reader could otherwise infer a designed asymmetry from their differing values.

A real-classifier client exists but is not used here. The accompanying repository also implements a structured-output client against a commercial LLM API (low-temperature decoding, schema-validated output), selectable from the command line as a drop-in replacement for the mock classifier. No result in this paper uses it: every number reported here comes from the mock classifier described above, whose only randomness is the single Bernoulli draw in Equation 7, with no language-model decoding step for a temperature or a schema to govern. We flag this plainly because the two clients share an interface, and a reader should not infer that any of this paper's numbers were produced by, or tuned against, a real language model.

3.5 Validation and study samples

We split the 4,853 household-waves into a 20% validation sample ($n = 970$) used only to estimate the screen's TPR and TNR, and an 80% study sample ($n = 3{,}883$) used for all three regressions in Section 4. Table 1 reports the full validation breakdown: accuracy 0.797, Cohen's $\kappa = 0.548$, and F1 = 0.700, alongside the confusion matrix each rate is computed from – we report the counts directly, not only the derived rates, so that TPR and TNR (the quantities Equations 5–6 actually consume) are independently verifiable rather than taken on faith. This is realistically imperfect, not an idealized round number, since it is drawn from the same finite-sample randomness as everything else in this design.

Table 1: Validation-sample classification performance ($n = 970$)

Metric	Value
Accuracy	0.797
Cohen's κ	0.548
F1	0.700
TPR (sensitivity)	0.762
TNR (specificity)	0.813
Confusion matrix: TP=230, FP=125, FN=72, TN=543	

4 Empirical Strategy

With D^* and D as defined above, we estimate three regressions of mean HAZ on a binary poverty indicator, all on the study sample:

1. **Benchmark**: HAZ regressed on the true (consumption-based) poverty indicator D^*. This is the best available estimate of the poverty-HAZ association in this sample, not

a known ground truth – it is itself subject to ordinary sampling error, a distinction that matters for interpretation and that the two synthetic companion pipelines in this portfolio, where the "true" effect is a planted constant, cannot illustrate.

2. **Naive**: HAZ regressed on the screened poverty indicator D directly, as an analyst without access to consumption data would be forced to do.

3. **Corrected**: the Rogan-Gladen coefficient correction (Equation 6), using $\widehat{\text{TPR}}$ and $\widehat{\text{TNR}}$ estimated once from the validation sample and held fixed thereafter.

All regressions use a dependency-free ordinary-least-squares solver, cross-validated against `statsmodels` on synthetic data in the accompanying repository's test suite, chosen for portability in restricted computing environments rather than for any methodological reason.

Bootstrap inference for the corrected estimator, and what it does not cover. Because $\hat{\beta}_1$ in Equation 6 is a nonlinear function of $\hat{q}$, which is itself nonlinear in $\widehat{\text{TPR}}$ and $\widehat{\text{TNR}}$, no closed-form standard error is available; we use a nonparametric bootstrap over the study sample. Concretely, with $\widehat{\text{TPR}}$, $\widehat{\text{TNR}}$ fixed at their validation-sample values throughout:

1. For $b = 1, \ldots, B$ (with $B = 300$, chosen after a convergence check reported below): draw a resample $\{(D_i^{(b)}, Y_i^{(b)})\}_{i=1}^{n}$ of size n with replacement from the study sample.

2. Compute $q^{*(b)} = \frac{1}{n}\sum_i D_i^{(b)}$, then $\hat{q}^{(b)}$ via Equation 5 and $\hat{\beta}_1^{(b)}$ via Equation 6, using the *same* $\widehat{\text{TPR}}, \widehat{\text{TNR}}$ in every b.

3. Report $\widehat{\text{SE}}(\hat{\beta}_1) = \text{sd}\big(\{\hat{\beta}_1^{(b)}\}_{b=1}^{B}\big)$ and a 95% confidence interval as the $[2.5, 97.5]$ percentiles of $\{\hat{\beta}_1^{(b)}\}_{b=1}^{B}$.

Because $\widehat{\text{TPR}}$ and $\widehat{\text{TNR}}$ are constants across every bootstrap draw by construction, this procedure estimates only the component of $\text{Var}(\hat{\beta}_1)$ arising from study-sample resampling; it is silent on the additional variance contributed by the fact that $\widehat{\text{TPR}}$ and $\widehat{\text{TNR}}$ are themselves estimates from a *finite validation sample*, not known constants. A more complete procedure would resample the validation and study samples *jointly* in each draw b, re-estimating $\widehat{\mathrm{TPR}}^{(b)}, \widehat{\mathrm{TNR}}^{(b)}$ alongside $\hat{\beta}_1^{(b)}$; we do not implement this here; Section 5.2 shows why the omitted variance component is not a minor rounding correction but, in a specific and characterizable regime, the dominant term.

We chose $B = 300$ after a direct convergence check on the main specification, sweeping the number of bootstrap replicates from 30 to 2,000. The bootstrap SE of $\hat{\beta}_1$ is 0.112 at $B = 30$ and 0.105 at $B = 50$, then settles into a narrow band between 0.102 and 0.109 from $B = 100$ through $B = 2{,}000$, with no further systematic drift beyond $B \approx 100$. $B = 300$ sits inside this converged band, not at its edge.

The correction's validity rests on non-differential misclassification: the screen's error must not depend on HAZ conditional on true poverty status. This holds by construction in our simulated screen (the mock classifier's flip probability depends only on the true label, and the field note never contains HAZ), but would require independent justification for a real deployed screen – for instance, if a real LLM's classification accuracy varied with household characteristics that are themselves correlated with child health (e.g., literacy, or urban/rural status), the correction as stated here would not apply without modification.

5 Results

5.1 Headline result

Table 2 and Figure 1 report the three estimates. The naive estimator attenuates the benchmark poverty-HAZ coefficient by 43% (from -0.357 to -0.204); the corrected estimator (-0.366) closes 94% of that gap and falls within the benchmark's 95% confidence interval.

To put -0.357 in substantive terms: against the study sample's overall HAZ standard deviation of 1.888, this is a gap of roughly 0.19 standard deviations between poor and non-poor household-waves. In levels, poor household-waves average a HAZ of -1.694 against

-1.337 for non-poor – a difference that also shows up in WHO stunting thresholds (WHO Multicentre Growth Reference Study Group, 2006): 46.0% of poor household-waves fall below HAZ $= -2$ (moderate-or-worse stunting) versus 36.5% of non-poor, a 9.5 percentage-point gap, and 25.3% versus 17.5% fall below HAZ $= -3$ (severe stunting), a 7.8 percentage-point gap. We report these as descriptive context for the coefficient's magnitude, not as a claim that poverty causes the gap in either direction – see the confounding discussion in Section 2.2 and the limitations in Section 6.

Table 2: Association between poverty and mean household HAZ, by estimator

Model	$\hat{\beta}_1$	SE	95% CI
Benchmark (true poverty)	-0.357	0.064	$[-0.482,\ -0.232]$
Naive (screened poverty)	-0.204	0.062	$[-0.325,\ -0.083]$
Corrected (matrix method)	-0.366	0.109	$[-0.542,\ -0.138]$

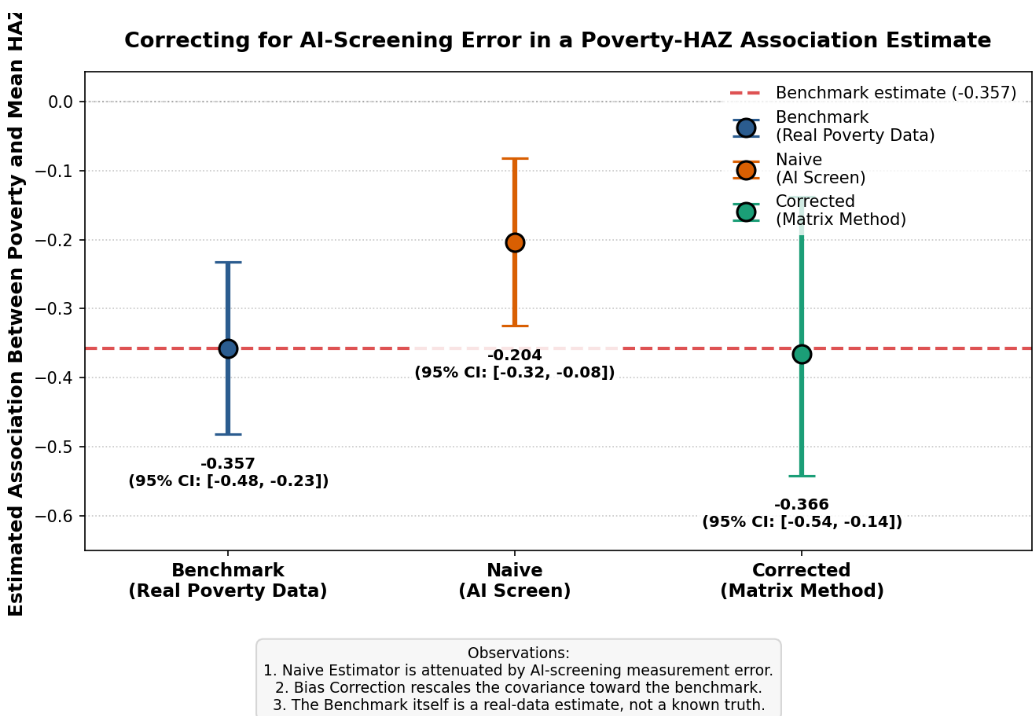

Figure 1: Point estimates and 95% confidence intervals for the three estimators, on the real ESS study sample ($n = 3{,}883$).

5.2 Sensitivity to the validation-sample size: an instability, not just a wider interval

The corrected estimator's own bootstrap SE grows as the validation sample shrinks – unsurprising, since smaller samples yield less precise TPR and TNR estimates. What is not obvious from the correction's algebra alone, and would not appear in the two synthetic companion pipelines' typical validation sizes (200–1,000 examples), is how the correction behaves as the *point estimate* of classifier skill (TPR+TNR−1) happens to land close to zero by chance in a small sample. We fix the true screen accuracy at 80% throughout – so the population value of TPR+TNR−1 is approximately 0.6 in every run – and draw 20 independent validation splits of $n = 38$ each (roughly the size a small-budget program might realistically afford for a validation subsample), re-estimating TPR, TNR, and the corrected coefficient in each.

Nineteen of twenty splits produce a corrected estimate between −0.16 and −0.59 – noisy, given the small sample, but recognizably in the vicinity of the benchmark (−0.29 in this particular subsample of the data used for this exercise). The twentieth split happens to estimate TPR$= 0.40$, TNR$= 0.71$ – both plausible draws individually, but combining to an estimated skill of $\text{TPR} + \text{TNR} - 1 = 0.114$, small enough that Equation 6's denominator amplifies ordinary covariance noise into a corrected coefficient of −34,545 (bootstrap SE 16,860) – five orders of magnitude from any sensible value, despite arising from the same true 80%-accurate screen as every other split. Figure 2 shows this directly: panel (a) is the full range including the divergent split; panel (b) excludes it and shows that, in the range where skill is estimated above roughly 0.2, corrected estimates cluster sensibly around the benchmark.

Why this happens: tracing the mechanism, not just the outcome. It would be unsatisfying to report a five-order-of-magnitude divergence and call it "noise" without explaining the channel. Decomposing the divergent draw's calculation term by term shows it is

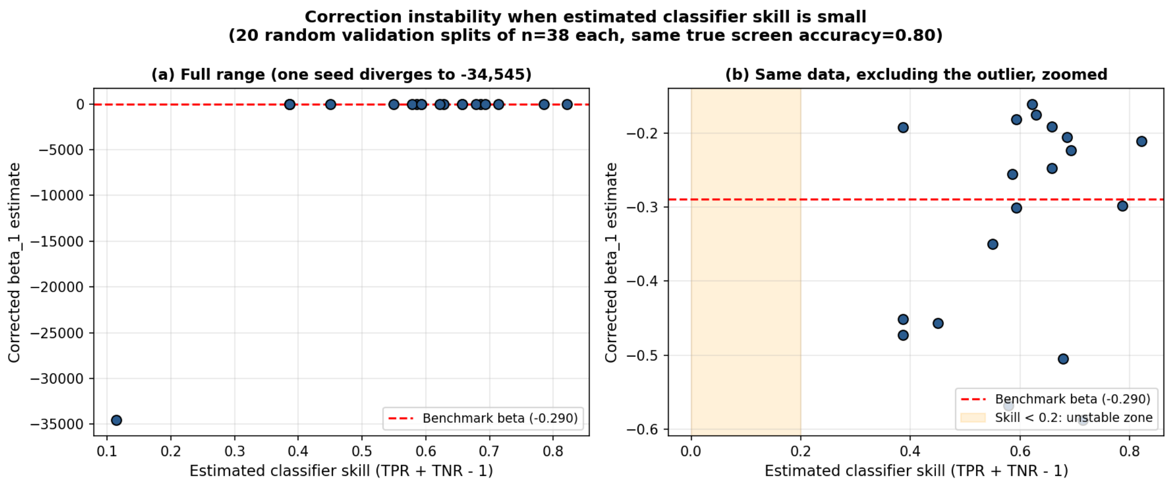

Figure 2: Corrected $\hat{\beta}_1$ against estimated classifier skill (TPR+TNR−1), across 20 independent validation splits of $n = 38$, all drawn from the same 80%-accuracy screen.

not a single blowup but a *compounding of two*. First, $\hat{\kappa} = 0.114$ is small, which alone inflates Equation 6's denominator. Second, and more consequentially, that same small $\hat{\kappa}$ appears in the denominator of the prevalence correction (Equation 5): $\hat{q} = (q^* - \widehat{\mathrm{FPR}})/\hat{\kappa}$ amplifies any noise in $q^* - \widehat{\mathrm{FPR}}$ by a factor of $1/\hat{\kappa}$, and in the divergent draw this pushed the raw prevalence estimate to $\hat{q}_{\mathrm{raw}} = 1.060$ – outside $[0, 1]$, as an estimated probability cannot be, and clipped by construction to $\hat{q} = 0.99999$. At that boundary, $\hat{q}(1 - \hat{q}) \approx 1.0 \times 10^{-5}$, collapsing Equation 6's denominator far further than $\hat{\kappa}$ alone would: $\hat{\kappa} \cdot \hat{q}(1 - \hat{q}) \approx 1.1 \times 10^{-6}$, against a typical well-behaved value near $0.6 \times 0.22 \approx 0.13$ – a five-order-of-magnitude collapse in the denominator that maps directly onto the five-order-of-magnitude divergence in $\hat{\beta}_1$. The instability is therefore not merely "$\hat{\kappa}$ is small"; it is that a small $\hat{\kappa}$ destabilizes $\hat{q}$ *first* (a standard delta-method argument gives $\mathrm{Var}(\hat{q}_{\mathrm{raw}}) = O(\hat{\kappa}^{-2})$ holding the numerator's sampling variance fixed, which we confirm by direct Monte Carlo simulation of Equation 5 across a range of κ), and the destabilized, boundary-clipped $\hat{q}$ then compounds multiplicatively with the same small $\hat{\kappa}$ in Equation 6.

This structure is a specific instance of a broader, well-studied econometric pathology: Equation 6 is algebraically a Wald-type ratio estimator, exactly the form taken by a just-identified instrumental-variables estimator ($\hat{\beta}_{\mathrm{IV}} = \mathrm{Cov}(Z, Y)/\mathrm{Cov}(Z, X)$), with $\hat{\kappa}$ playing the role of a first-stage coefficient. The weak-instruments literature (Staiger and Stock, 1997) establishes that such ratio estimators become badly behaved – non-normal, fat-tailed, and unreliable exactly when the denominator's population value is small relative to its own sampling variance, independent of how large the outcome sample itself is. Equation 6 inherits this pathology in full: a large study sample ($n = 3{,}883$ here) does nothing to stabilize $\hat{\beta}_1$ if $\hat{\kappa}$ comes from a small validation sample, because it is $\hat{\kappa}$'s *own* sampling variance, not the study sample's size, that determines whether Equation 6 is well-identified. The broader kinship between instrumental-variable estimators and corrections for classical measurement error is well established – Bound et al. (2001) survey IV-based fixes for mismeasured regressors at length – and a validation-sample-based correction for a mismeasured binary regressor is, structurally, one member of that family. What we have not seen made explicit specifically for the Rogan-Gladen coefficient correction is the weak-instrument *diagnosis*: that its known instability at small $\hat{\kappa}$ is not a generic small-sample-noise caveat but the identical failure mode formalized by Staiger and Stock (1997), with the same practical remedy (a pre-estimation strength check on $\hat{\kappa}$, rather than a post-hoc adjustment to the point estimate).

We did not set out to demonstrate this instability; it appeared during a systematic sweep of validation-sample sizes intended to characterize ordinary standard-error scaling, and a one-in-twenty failure rate is large enough that we consider it the paper's more actionable finding. Guided by the weak-instrument analogy above – where the standard practical remedy is a pre-estimation strength test (e.g., the first-stage F-statistic rule of thumb) rather than a post-hoc correction to the point estimate – we propose, provisionally, that a validation sample should be large enough to be confident the estimated skill exceeds some conservative threshold (we suggest 0.2–0.3 as a starting point, pending further calibration) *before* the point correction is reported, and that the bootstrap should more properly resample the validation split jointly with the study split – which our implementation, and to our knowledge every standard exposition of this correction, does not do (see Section 6). The $O(\hat{\kappa}^{-2})$ result above gives this threshold an analytic starting point rather than leaving it purely simulated, though deriving its precise finite-sample value as a function of both the validation and study sample sizes remains open.

5.3 Sensitivity to screen accuracy

Table 3 sweeps the target screen accuracy a from 65% to 95%, holding the validation split at its default size. The TPR and TNR columns are, as in Table 1, single realized validation-sample estimates at each level (visibly unequal within a row, e.g. 0.609 vs. 0.653 at $a = 0.65$), not the population values – which Equation 7 establishes are exactly equal to a at every level by the mock classifier's construction. Attenuation falls monotonically as accuracy rises (from 79% attenuation at 65% accuracy to 7% at 95%), as expected. The correction closes 70–95% of the gap across most of this range; at 95% accuracy, where the naive gap is already small, the reported "percent of gap closed" becomes an unstable statistic (dividing by a small naive gap), which we note as a definitional caveat rather than a substantive finding.

This 65–95% range is not an arbitrary bracket: it brackets the real accuracies Gilardi et al. (2023) report for zero-shot LLM classification on binary-relevance annotation tasks most comparable to ours in structure – 59% and 70% on two content-moderation tweet-classification tasks, 81% on a content-moderation news task, and 83% on a US Congress tweet-classification task, i.e., a real reported range of roughly 59–83% for tasks of this general kind, against which our 80% default and 65–95% sweep are a reasonable if imperfect proxy rather than an arbitrary convenience choice. It is also consistent with the broader finding, from a different but related screening technology, that fielded proxy means tests routinely misclassify a substantial share of households: Brown et al. (2018) report that standard PMTs achieve weak coverage of the genuinely poor and high exclusion-error rates across the nine African countries they study, without landing on a single comparable accuracy figure – non-trivial misclassification of the kind this paper's correction is designed for is the empirical norm in this literature, not a pessimistic assumption of ours.

Table 3: Attenuation and correction performance across screen accuracy levels

Accuracy	TPR	TNR	Naive $\hat{\beta}_1$	Corrected $\hat{\beta}_1$	Attenuation
0.65	0.609	0.653	−0.076	−0.305	79%
0.70	0.639	0.699	−0.136	−0.419	62%
0.75	0.699	0.747	−0.171	−0.406	52%
0.80	0.762	0.813	−0.204	−0.366	43%
0.85	0.808	0.855	−0.227	−0.350	37%
0.90	0.864	0.904	−0.275	−0.364	23%
0.95	0.940	0.952	−0.334	−0.380	7%

5.4 Robustness to the poverty-line definition

Table 4 varies the within-wave poverty threshold from the bottom quintile to the median. The benchmark, naive, and corrected estimates all remain in a consistent range across this variation, and the correction closes a majority of the attenuation gap at every threshold tested (59–97%), indicating the headline result is not an artifact of the particular one-third cutoff chosen as the main specification.

Table 4: Robustness to the within-wave poverty threshold

Threshold	Benchmark $\hat{\beta}_1$	Naive $\hat{\beta}_1$	Corrected $\hat{\beta}_1$	Gap closed
Bottom 20%	−0.464	−0.153	−0.337	59%
Bottom 25%	−0.382	−0.167	−0.337	79%
Bottom 33% (main)	−0.357	−0.204	−0.366	94%
Bottom 40%	−0.373	−0.220	−0.378	97%
Bottom 50%	−0.364	−0.188	−0.322	76%

6 Discussion

6.1 What a practitioner should take from this

If a program is considering an AI or LLM-based poverty screen as a stand-in for full consumption data, this paper's exercise suggests two things worth checking before trusting a corrected estimate: first, whether the assumption of non-differential misclassification is plausible for the actual screen and outcome in question (not assumed by construction, as it is here); and second – more concretely actionable – whether the validation sample used to estimate TPR and TNR is large enough that the estimated skill $\text{TPR} + \text{TNR} - 1$ is comfortably bounded away from zero, not merely positive. Our simulated instability (Section 5.2) suggests reporting a confidence interval on the skill estimate itself alongside the corrected coefficient, and treating a corrected estimate as unreliable – regardless of how confident the point estimate looks – whenever that interval comes close to including zero.

6.2 Relation to the companion pipelines

This paper is one of three related demonstrations of the same correction in this portfolio. The other two apply it to a simulated financial-sentiment classification task and to a fully synthetic outcome with a planted true effect; both are useful for validating that the correction recovers a *known* answer under controlled conditions, but neither can speak to what happens when the "true" value being corrected toward is itself an estimate with sampling error, nor – since both use larger, round validation samples in their default configurations – did either surface the small-sample instability documented here. We view this as evidence that testing a method against real, independently-measured data, even when the "treatment" being measured is itself simulated, surfaces failure modes that a fully synthetic demonstration is structurally unlikely to expose by design.

6.3 Limitations

We restate these plainly rather than leaving them implicit. (1) The screen is a controlled simulation, not a real LLM classification; no claim is made about what accuracy a real deployment would achieve on real household descriptions. (2) The field notes are constructed from real covariates but are not real enumerator text; the ESS survey has no free-text field. (3) The within-wave relative poverty definition is a modeling choice, not an official poverty line. (4) The bootstrap resamples the study sample only; it does not propagate sampling uncertainty in the validation-sample-derived TPR/TNR, which Section 5.2 shows can be the dominant source of instability – a joint resampling scheme is a natural extension we have not implemented. (5) The proposed skill threshold (Section 5.2) is still provisional: the $O(\hat{\kappa}^{-2})$ variance result and the weak-instruments analogy explain *why* the instability occurs and give the threshold an analytic rather than purely simulated basis, but we have not derived its precise finite-sample value as a joint function of validation-sample size, study-sample size, and the clipping boundary that compounds it – that derivation, plausibly adaptable from the weak-instrument pre-testing literature, is left for future work. (6) As Section 2.2 discusses explicitly, none of the three estimators in Table 2 – including the benchmark, which uses real, unscreened poverty status – adjusts for confounders such as geographic remoteness that plausibly affect both household poverty and child HAZ directly, nor for reverse causality running from child health back to measured household consumption; the measurement-error correction this paper studies and the separate problem of causal identification are independent, and solving the former does not touch the latter.

7 Conclusion

Using real consumption and child-health data from the Ethiopia Socioeconomic Survey, we show that a simulated AI poverty screen of realistic accuracy would attenuate a poverty-health regression by 43%, and that a standard measurement-error correction recovers 94% of the resulting bias using only a validation subsample. We also show, empirically rather than asserting from theory, that this correction is not reliable at every validation-sample size: at $n = 38$, a plausible small-survey validation size, one in twenty random draws produced a corrected estimate that diverged by five orders of magnitude despite an unchanged, well-behaved underlying classifier. We propose that any application of this correction report the estimated classifier skill (TPR+TNR−1) with its own uncertainty, and treat the correction as unreliable when that quantity is not clearly bounded away from zero – a check we consider at least as important as reporting the corrected point estimate itself.

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